

Reihaneh Teimouri

Machine Learning Applied Scientist

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About

Machine Learning Applied Scientist with 3+ years of industry experience in medical imaging AI, generative models, and deep learning. Specialized in diffusion models, LLM, agentic AI, and foundation models, from research to production deployment. M.Sc. Computer Science, MICCAI 2024 · McGill Surgical Innovation Fellow.

Work Experience

Apr 2026 - Senior Machine Learning Applied Scientist, Prenuvo

- Present
 - Leading development of a unified algorithmic framework for whole-body MRI analysis and building agentic AI workflows using LLMs, LangChain, and LangGraph.
 - Researching unsupervised lesion detection methods using backbone and foundation models.
 - Contributing to body-composition MRI models for personalized and explainable health insights for patients.

Apr 2025 - Machine Learning Applied Scientist, Prenuvo

- Apr 2026
 - Contributed to age estimation models and AI-generated whole-body composition reporting system.
 - Developed lesion detection models across multiple MRI series.

Sep 2024 - Machine Learning Scientist Intern, Prenuvo

- Apr 2025
 - Contributed to FDA 510(k) submission for AI body composition reporting system.
 - Developed Diffusion models for personalized health reports from whole-body MRI data.

2022 - 2023 Machine Learning Scientist Intern, Synthesis Health

- Developed and trained machine learning models for segmentation and registration of medical images.

Education

2021–2024 M.Sc. in Computer Science, *Concordia University*, Montreal, Canada, **GPA: 4.0/4.0**

Selected Courses: Neuroimage Computing, Deep Learning, Computer Vision

2021–2023 NSERC Surgical Innovation Program Fellow, *McGill University*, Montreal, Canada

Cross-disciplinary program combining engineering, computer science, business, and surgery.

2015–2020 B.Sc. in Software Engineering, *Isfahan University of Technology (IUT)*, Iran

Selected Courses: Multimedia Systems, Algorithm Design, Artificial Intelligence, Programming, DataBase

Professional skills

Programming Python, C++, Java, R, MATLAB, SQL, Bash

Frameworks PyTorch, MONAI, TensorFlow, Keras, Diffusers, Hugging Face, scikit-learn

Core AI Foundation Models, Generative AI (Diffusion Models, GAN, VAE), LLM, Transformers, Domain Adaptation, Self-Supervised Learning, Medical Image Analysis, GradCAM, Explainable AI (XAI)

Agentic AI AI Agents, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph

MLOps AWS (S3, EC2, SageMaker), GCP, Docker, ZenML, MLflow, Ray / Anyscale, Poetry, Git, CI/CD

Other Jira (Agile), Gradio, Streamlit, Linux System Administration

Publications

2024 **R. Teimouri**, Y. Xiao, M. Kersten. "CT-based brain ventricle segmentation via diffusion Schrödinger Bridge without target domain ground truths." Early acceptance at MICCAI2024.

2020 **R. Teimouri**, K. Mostafaie, Z. Nabizadeh, N. Karimi, S. Samavi. "Region of Interest Identification for Brain Tumors in Magnetic Resonance Images." International Conference on Electrical Engineering, Iran.

2020 **R. Teimouri**, Z. Nabizadeh, N. Karimi, S. Samavi. "An Abstraction Model for Semantic Segmentation Algorithms." International Conference on Machine Vision and Image Processing.